1.Understanding Survey Scale

A sonar survey can produce a very large amount of data.A single image is not the main problem. The problem comes when thousands of images are produced during a long survey.For example, DRISHTI uses image tiles of:640 × 640 pixels.The sonar survey is divided into many such tiles.Each tile is processed separately by the detection model.Therefore, a large survey can be viewed as:Large sonar survey → many tiles → AI inference on each tile → many detection results.The number of tiles depends on factors such as:Survey area,sonar resolution,swath width,tile size,overlap between tiles,survey speed.

Lever	Current	At 10×	How we verify
Storage	Local/object storage	Scalable object storage	Measure storage
Processing	4 workers ≈40 tiles/s	Add workers	Benchmark 1/2/4/8/...
Dataset	6,405 tiles	10× expanded dataset	Count tiles/classes
Difficult classes	Limited real data	More real wreck/cylinder	Per-class AP
Geography	Current regions	New regions	Cross-region test
Review	Single queue	Multiple analysts	Review time
Deployment	FP32 ONNX	Same/lightweight pipeline	Hardware benchmark

DRISHTI can be scaled in six main ways. First, throughput can be increased by adding more worker processes and CPU cores so that more sonar tiles can be processed at the same time. Second, survey size can be increased because each sonar tile can be processed independently, which makes large surveys suitable for parallel processing. Third, storage can be scaled by keeping large sonar images in edge or object storage while storing only small detection records in the database. Fourth, new hazard classes can be added by collecting labelled data and retraining the model without making major changes to the system architecture. Fifth, deployment targets can be increased because the same lightweight inference code can run on different platforms such as a laptop, server, or Jetson device. Finally, geographic coverage can be expanded by fine-tuning the model with confirmed detections from new regions. In this way, DRISHTI can grow in terms of processing speed, survey area, storage, hazard types, hardware, and geographical regions.

If the DRISHTI dataset becomes 10× larger, we would scale the system in several ways without changing its basic architecture. First, we would use scalable object storage to store the much larger sonar imagery. Second, we would increase the number of processing workers so that more sonar tiles can be processed in parallel. Third, we would retrain the YOLOv8s model using the additional data to improve its detection performance. Fourth, we would collect more real examples of difficult classes such as shipwrecks and mine cylinders. Fifth, we would add more hard-negative images to reduce false detections. Sixth, we would include data from different survey regions and sonar conditions so that the model becomes more robust to geographical and environmental changes. Seventh, we would maintain the same lightweight inference pipeline so that it can continue running on suitable edge or server hardware. Eighth, we would scale the database and storage system so that detection records and metadata can be handled efficiently as the number of findings increases. Ninth, we would expand the analyst review system so that multiple analysts can review detections and medium-confidence results can be prioritized. Finally, we would monitor the model's performance on the new data and fine-tune it whenever new regions, sonar conditions, or hazard types introduce new errors. Therefore, at 10× data, the main changes would be in storage, processing capacity, training data, model robustness, and review capacity rather than completely redesigning DRISHTI.

2.Measured assumed and next

MEASURED

These are values that have already been tested or obtained from the current DRISHTI implementation.The current system processes sonar imagery by dividing it into independent 640×640 tiles. On CPU, the lightweight FP32 ONNX model takes approximately 90 ms per tile for inference. A single worker can process roughly 11 tiles per second, while four workers can reach approximately 40 tiles per second because multiple tiles can be processed simultaneously. The current survey-scale estimate is around 5,100 tiles per day,under the current planning assumptions, the estimated processing-equivalent survey coverage is around 130 hectares per day.Scaling InsightUnder the current assumptions, the AI inference stage has substantially more capacity than the estimated daily tile production. Therefore, at larger scale the main bottleneck is expected to shift from model inference toward data I/O, geotagging and especially analyst review.This is why the 10× experiment will measure the complete pipeline ratherthan inference speed alone.
The torch-free inference worker requires roughly 250 MB of runtime memory, while the complete lightweight runtime is around 210 MB.The approximately 210 MB value refers to the lightweight installed runtime environment, while approximately 250 MB refers to the memory used by an active inference worker during execution.
The FP32 ONNX model is approximately 44.8 MB. These measurements show that the current inference pipeline is already lightweight and that increasing the number of workers is a practical way to increase processing throughput. The current model also has a measured mAP@50 of 0.641, with precision of approximately 0.734 and recall of approximately 0.629. These measured results provide the baseline against which the 10× data version can be compared.

ASSUMED / PLANNED

These are not measurements from a completed 10× experiment. They are engineering plans for how DRISHTI would be scaled if the amount of sonar data becomes approximately ten times larger.At 10× data, we would first move the larger sonar imagery to scalable object storage instead of depending on a single local machine. Since the sonar tiles are independent, the processing workload can be divided among a larger number of workers. For example, if four workers currently provide around 40 tiles per second, additional workers could be added when more CPU resources are available. The exact throughput at 10× data would need to be measured rather than assumed. More examples would be collected for difficult classes such as shipwrecks and mine cylinders, while additional hard-negative examples would be included to help reduce false positives.Examples would be collected for difficult classes such as shipwrecks and mine cylinders, while additional hard-negative examples would be included to help reduce false positives. Data from different geographical regions, sonar conditions, resolutions, and survey environments would also be added so that the model does not depend too strongly on one type of sonar imagery.Storage would be scaled separately from the database. Large raw sonar files would remain in object or edge storage, while the database would mainly contain smaller detection records such as class, confidence, coordinates, dimensions, ping information, and timestamps. This prevents database size from growing at the same rate as the raw sonar imagery.The analyst-review system would also need to scale. A 10× increase in input data can produce a much larger number of candidate detections, so the system should prioritize medium-confidence detections and allow multiple analysts to work on the review queue. The goal would not simply be to increase AI inference speed, but to make sure the complete pipeline—from sonar input to verified detection—can handle the larger workload.

The same lightweight deployment approach would be retained. The existing torch-free ONNX Runtime pipeline could continue to be used on servers or edge devices, with suitable hardware selected according to the required throughput. New hazard classes could also be introduced later by collecting labelled examples and retraining the model rather than redesigning the complete system. Geographic expansion could similarly be handled through fine-tuning with confirmed detections from new regions.

NEXT

The next step would be to create an actual **10× data experiment** instead of only estimating its behaviour. First, we would define the target 10× dataset and record exactly how many images, tiles, classes, and regions it contains. Second, we would place the larger dataset in scalable storage and organize it so that training and inference data can be accessed efficiently.Third, we would run the existing model on the larger dataset to establish a baseline. We would measure total processing time, tiles processed per second, CPU and memory usage, storage requirements, and the number of detections generated. Fourth, we would increase the number of workers and measure how throughput changes with 1, 2, 4, 8, and more workers. This would tell us the actual scaling limit instead of assuming that throughput increases linearly.Fifth, we would retrain the model using the expanded dataset and compare the new model with the current baseline using mAP@50, precision, recall, per-class performance, false-positive rate, and calibration error. Sixth, we would specifically test difficult classes and hard-negative cases to determine whether the larger dataset reduces false detections.
Seventh, we would test the model on regions that were not present in the training data. This would help measure whether the model generalizes to new geographical and sonar conditions. Eighth, we would test the complete edge pipeline on the intended hardware and measure real inference time, memory usage, storage consumption, and offline operation.Finally, we would measure the analyst-review workload. If the AI becomes faster but generates too many candidates for humans to review, the bottleneck has simply moved from computation to human review. Therefore, the final 10× experiment should measure the entire pipeline rather than only model inference. The result would give us a real basis for deciding how much hardware, storage, training data, and analyst capacity are required for larger operational surveys.

3.Survey

Estimated Daily Survey Capacity:-For the current planning setup, the estimated survey coverage is around:130 hectares per day At approximately 5 cm per pixel and 50% overlap, this can produce around:5,100 tiles per day.These numbers are planning estimates for the current system and should not be treated as guaranteed field performance.The actual number of tiles can change depending on the sonar settings and survey conditions.The important point is that DRISHTI is designed to handle thousands of image tiles rather than only a few example images.

Estimated AI Inference Processing Time:-The AI model has been tested on individual 640 × 640 tiles.The FP32 ONNX model takes approximately:90 milliseconds per tile on the tested CPUThis means the model can process a large number of tiles without requiring a very large server.For example, the project measurements estimated approximately:7.7 minutes for one processing worker for around 5,100 tiles under the tested processing assumptions.With four workers, the estimated time was around:2.1 minutesAgain, these are processing estimates and not the complete time required for a real survey.
These values represent estimated AI inference time only. They do not represent total survey processing time, which also includes file I/O, preprocessing, geotagging, database operations, human review and report generation.

Inference Speed :-Underwater surveys can produce a large amount of data.If every image takes a long time to process, the processing backlog can become very large.For this reason, the system was designed around a relatively small detector.YOLOv8s was selected instead of using a much larger model.The goal was to find a balance between:Accuracy,speed,model size,hardware requirements.The measured CPU inference time of around 90 ms per tile shows that the current model is fast enough for the planned processing pipeline.The system can also use GPU hardware for higher processing capacity.

Sensor Processing Rate:-The system also considers how quickly new sonar data may arrive.For the current engineering calculation, a processing cadence of:5 Hz was assumed.This means one processing unit arrives approximately every:200 millisecondsThe system therefore uses:200 ms as the hard processing limit and:100 ms as the engineering targetThe measured FP32 ONNX processing time of approximately 90 ms per tile is within this 100 ms target on the tested development CPU.This gives some extra room for other processing tasks.However, the 5 Hz value is an engineering assumption, not a verified output rate of the EdgeTech 2205 sonar.The actual sensor rate should be measured during hardware integration.
NOTE:- The 5 Hz processing cadence is an engineering assumption, not a verified output rate of the EdgeTech 2205. The actual sensor output rate will be measured during hardware integration.

CPU Results:-The system was able to achieve approximately 90 ms per tile on a CPU.This is useful because it means the system does not completely depend on a powerful GPU to operate.The planned Jetson Orin NX can provide additional processing headroom.The Jetson is therefore mainly useful for:Faster processing,better power efficiency,running multiple operations,handling future improvements,supporting real-time or near-real-time operation.A CPU fallback can also be useful if the GPU becomes unavailable.

Large-Scale Processing:-When the survey becomes larger, the number of tiles increases.For example:1 survey → 5,100 tiles.A larger survey may produce:10,000+ tiles.The same basic processing pipeline can continue to work.The main change is that more workers may be required.The system can divide the workload between multiple workers.This is called parallel processing.It reduces the total processing time when enough computing resources are available.

Human Review at Scale:-Human review is especially important for medium-confidence detections.The current confidence system divides results into three groups:80% or aboveThese detections can be automatically confirmed.30% to 80%These detections are sent to a human review queue.Below 30%These detections are normally dropped.This approach reduces the amount of manual work.Instead of an analyst checking every sonar tile, the analyst can focus on detections that are more likely to contain something important.However, if a large survey produces too many medium-confidence detections, the review workload can still become high.This means future improvements should focus not only on increasing model accuracy but also on reducing unnecessary alerts.

False Positives:-False positives are an important limitation in underwater sonar detection.A false positive happens when the model detects something as a hazard even though it is not actually the target.For example, a normal seabed structure may look similar to:A shipwreck,a pipeline,a cylinder,a fishing net.This is why DRISHTI uses several filtering steps.
The system uses:Background images,Hard negatives,confidence thresholds,per-class calibration,acoustic shadow checks,human reviewThese steps are designed to reduce unnecessary alerts.

Background Data:-Background data is especially important when working at large scale.The dataset includes object-free seabed images.These images help the model learn that not every bright or dark region is a hazard.Without enough background examples, the model may learn shortcuts.For example, it might associate a particular bright seabed pattern with a shipwreck.Adding background and hard-negative samples helps reduce this problem.This is important because real sonar surveys contain much more normal seabed than actual hazards.

4. Different Classes Have Different Limits

The model does not perform equally on every class.The final test results showed:Submarine pipeline: very strong performance,ghost net: high test performance, but mainly based on synthetic data,mine cylinder: moderate performance and shipwreck is more difficult.
This difference is expected because the objects have different shapes and visual characteristics.

Pipeline:-Pipelines are often long and relatively regular.This makes them easier for the detector to identify.

Shipwreck:-Shipwrecks can have many different shapes and orientations.The available real labelled data was also limited.

Mine Cylinder:-Cylinders can look similar to other underwater objects.Their appearance can change depending on orientation and sonar conditions.

Ghost Net:-Ghost nets can be thin, weak and fragmented in sonar images.The main limitation is that the current ghost-net training data contains a large synthetic component.Therefore, the ghost-net test score should not be treated as real-world field accuracy.

5. Limitation

Dataset Size Limitation

The amount of labelled data is one of the biggest limitations of DRISHTI.The final dataset contains:**6,405 image tiles** with:4,775 training tiles,780 validation tiles and 850 test tiles.However, the number of real examples is not equal for every class.Some classes have much more useful real data than others.This creates an imbalance in how well the model can learn each hazard.Simply adding more images does not always solve the problem.The additional images need to represent the conditions that the model will see in real surveys.

Synthetic Data Limitation

Synthetic data was used mainly for ghost-net training because suitable labelled real ghost-net data was not available in the required form.Synthetic data is useful because it allows the team to create more examples.However, synthetic sonar images may not perfectly match real sonar images.Real sonar data can contain:Different noise,Different brightness,Different sea conditions,Different object shapes,Different sonar frequencies,Different viewing anglesTherefore, a model that performs well on synthetic test data may not perform equally well on real field data.For ghost nets, real-world validation is especially important.

Geographic Generalisation

Another limitation is geographic generalisation.A model trained on sonar data from one region may not work equally well in another region.This can happen because seabed conditions change.For example:Sand looks different from rock,water conditions can change,sonar settings can change,object appearance can change,background textures can change.Therefore, a model should ideally be tested on geographically different data.If performance drops in a new region, some additional fine-tuning may be required.

Location Accuracy Limitation

The system can calculate a geographical position for a detected object.However, the final position should not be treated as an exact point.The tow-fish navigation position itself can be very accurate.But the detection position also depends on:Image scaleAcross-track positionSlant-range conversion,Vehicle motion,Navigation information,sonar geometryDuring testing, two navigation paths placed the same target approximately:122 metres apartThis shows that the final detection location can have a much larger uncertainty than the navigation fix itself.Therefore, DRISHTI should use an uncertainty radius or area.This is safer than showing a detection as an exact point.

Communication Limitation

Underwater communication is another major limitation.
An underwater vehicle cannot normally send large sonar images through an acoustic modem at a useful rate.Because of this, DRISHTI cannot depend on sending every sonar tile to a remote server while the vehicle is underwater.The system therefore follows an edge-processing approach.The vehicle processes the sonar data locally.Only small information such as:Object class,position,confidence,radius,ping ID may be transmitted during the mission.The full sonar data can be transferred after the vehicle returns.This limitation strongly affects the system architecture.

Storage Requirements

Large surveys also require significant storage.The reference AUV platform provides local mission storage.The system is designed to store the sonar data and detection results locally during the mission.This is important because the vehicle cannot depend on continuous communication with the surface.The system therefore follows:
Collect → Process → Store locally → Recover vehicle → Transfer data
This makes the system more practical for long underwater missions.

Edge Hardware Limitation

The planned edge computer is the:NVIDIA Jetson Orin NX 16 GB.The system targets FP16 and TensorRT for the Jetson.However, the actual FP16/TensorRT performance on the final Jetson hardware still needs to be measured.The current measured 90 ms result is from the FP32 ONNX model on the development CPU.Therefore, it would be incorrect to claim a final Jetson FPS or latency before running the system on the actual device.This is an important difference between:Measured result and Future deployment target.

INT8 Limitation

INT8 quantization was tested to reduce model size.The INT8 model was approximately:11.5 MBwhich is smaller than the FP32 model.However, the tested INT8 version produced approximately:0 precision and 0 recall on the evaluation.It also took around:164 ms per tile on the tested CPU.Therefore, the INT8 version was not selected.This shows that a smaller model is not always a better model.The system chose the working FP32 version instead.FP16 remains a future deployment target for suitable GPU hardware.

Dashboard Limitation

The dashboard is an important part of the complete system, but it is not the same as the AI inference pipeline.The current dashboard uses a React and Leaflet-based structure.It is designed to support:Upload,map display,detection review,results,export.

However, the dashboard is still at the integration or wiring stage.Therefore, it should not be presented as a completely finished production system.The main AI, confidence and geotagging components are more mature than the complete user interface.

Public Map Limitations

A public debris map also creates additional problems.Not every detection should automatically become a public map marker.For example, shipwreck locations may have heritage or safety concerns.Possible mine or ordnance detections also need to be handled carefully.Therefore, a public map should show:Analyst-confirmed hazards onlyIt should also show an uncertainty area instead of an exact location.The map should be treated as an advisory system.It should not replace official maritime navigation warnings.

6. Current Scale Summary

The current system can be summarized using the following numbers:

Item	Current Value
Dataset size	6,405 tiles
Training tiles	4,775
Validation tiles	780
Test tiles	850
Tile size	640 × 640
Estimated daily survey area	~130 ha
Estimated tiles/day	~5,100
FP32 ONNX size	44.8 MB
FP32 CPU latency	~90 ms/tile
Torch-free runtime	~210 MB
Engineering target	≤100 ms
Assumed processing cadence	5 Hz
INT8 latency	~164 ms/tile
Detection location difference observed	~122 m

These values describe the current tested system and planning assumptions.

They should not all be treated as guaranteed field performance.